Notes: Frakes & Gruber (RES, 2026)

The Effect of Provider Diversity on Racial Health Disparities: Evidence from the Military

Contents

1 Big picture

- **Question.** Does increasing the racial diversity of the medical provider workforce reduce racial health disparities?
- **Policy object.** The paper studies the effect of a higher *community/provider-workforce share* of Black providers, not just the effect of one racially concordant physician-patient encounter.
- **Empirical setting.** The Military Health System (MHS) provides unusually rich claims data, broad geographic coverage, and provider demographic information because many military providers are themselves beneficiaries in the system.
- **Clinical focus.** The analysis studies chronic, deadly, but manageable conditions: diabetes, hypertension, hypercholesterolemia, and clinical atherosclerotic cardiovascular disease.
- **Why chronic disease matters.** Chronic disease management requires repeated interaction, communication, trust, medication adherence, diagnostic monitoring, and sustained care. These are exactly the channels through which provider diversity may affect outcomes.
- **Identification idea.** The authors use patient moves across military bases. Bases differ in the share of Black providers. Moving changes a patient's exposure to Black providers in a plausibly exogenous way.
- **Core specification.** The design compares Black versus non-Black patients, before versus after the move, and moves that increase versus decrease the Black-provider share.
- **Main preventive-care result.** A one-standard-deviation move-induced increase in the Black-provider share raises medication fill days for Black relative to non-Black patients by roughly 2.7–3.0 days.
- **Main mortality result.** The same increase reduces mortality for Black relative to non-Black patients by about 0.18–0.19 percentage points, roughly an 18% decline relative to the mean mortality rate.
- **Mechanism evidence.** Effects are strong in chronic-care and planned-decision settings, but not in emergency-room settings. This points toward trust, communication, and patient decision-making rather than purely provider-driven clinical intensity.
- **Spillover result.** The paper argues that roughly half or more of the effects may arise through spillovers onto non-Black providers or broader trust in the health system, not only through direct patient-provider racial concordance.
- **Mortality decomposition.** Improved preventive medication adherence explains about 31% of the estimated mortality effect; the remaining effect likely reflects other channels such as lifestyle, monitoring, trust, or unmeasured preventive behavior.
- **Economic takeaway.** Workforce diversity can be a health-production input. Increasing the share of Black providers can reduce racial health gaps even in a setting with guaranteed access to care.

2 Data and measurement

2.1 Military Health System data

- The MHS covers active-duty military, retirees, and dependents.
- The data contain claims for direct care and purchased care.
- Patients are assigned to bases according to the closest Military Treatment Facility catchment/service area.
- The paper focuses on patients who live within 10 miles of the base, making the local provider-race measure more relevant.
- The sample period is 2003–2013.

Interpretation. The MHS is useful because patients face a common insurance environment and the researcher can observe rich claims and demographic data. This reduces some access-to-insurance confounds that would arise in broader U.S. claims data.

2.2 Provider race

- Provider race is measured by exploiting the fact that active-duty and military-retiree providers are also MHS beneficiaries.
- The paper uses self-identified race fields and focuses mainly on Black versus non-Black providers.
- The key exposure is the base-level share of providers treating a given chronic condition who are Black.
- The key treatment variation is the move-induced change in this Black-provider share.

Interpretation. The provider-race measure is not simply individual concordance. It is a local-care-environment measure, capturing both direct access to Black providers and indirect spillovers through the local provider community.

2.3 Patient samples

- The main analysis uses adult MHS patients with diabetes, hypertension, hypercholesterolemia, or clinical atherosclerotic cardiovascular disease.
- The unit of observation is often a patient-by-disease-category cell.
- Patients must move once across bases during the sample period.
- The paper uses additional samples for biomarkers, emergency-room care, childbirth, and cancer screening.

Interpretation. Chronic-disease patients are a high-value sample for studying trust and communication because treatment quality depends heavily on long-run adherence, monitoring, and repeated clinical engagement.

2.4 Outcome variables

- **Preventive care:** medication fill days, especially guideline-relevant preventive medications.
- **Mortality:** incidence of mortality over the sample.
- **Biomarkers and symptoms:** systolic blood pressure and self-reported pain.
- **Mechanism outcomes:** ER intensity, planned and unplanned caesarean delivery, cancer screening.
- **Balance outcomes:** age, active-duty status, male share, RVUs, Charlson index, inpatient days, pay-grade, predicted mortality, and pre-move medication fills.

Interpretation. The outcome set is designed to distinguish between direct clinical intensity, patient trust/communication, preventive-care adherence, and downstream health consequences.

2.5 Treatment variable

Let P_j^B denote the share of relevant providers at base j who are Black. For a patient moving from origin base $o(i)$ to destination base $d(i)$, define

$$\Delta P_j^B = P_{d(i)}^B - P_{o(i)}^B.$$

The paper normalizes ΔP_j^B so that estimated coefficients are interpreted as the effect of a one-standard-deviation increase in the Black-provider share.

Interpretation. This scaling is important. A coefficient such as 2.676 fill days does not mean a one-percentage-point provider-share change; it means a one-standard-deviation move-induced change in local provider diversity.

3 Empirical specifications and interpretation

3.1 Triple-difference movers specification

The core estimating equation is:

$$\Delta Y_{ij} = \alpha + \beta B_i + \delta \Delta P_j^B + \tau B_i \times \Delta P_j^B + \Omega \Delta P_j^X + \gamma B_i \times \Delta P_j^X + \pi M_i + \Psi B_i \times M_i + \mu_j + \varepsilon_{ij}. \quad (1)$$

- ΔY_{ij} is the post-move minus pre-move change in the outcome.
- B_i is an indicator for Black patient.
- ΔP_j^B is the move-induced change in the base Black-provider share.
- ΔP_j^X captures move-induced changes in other base characteristics.
- M_i contains baseline patient characteristics.
- μ_j can include sending and receiving base fixed effects.
- τ is the coefficient of interest.

Interpretation. The design is a triple difference: Black versus non-Black patients, before versus after moving, and moves toward versus away from more Black-provider exposure.

3.2 Meaning of the key coefficient

The coefficient τ measures:

Effect on Black patients relative to non-Black patients of a move-induced increase in Black-provider share.

The absolute effect for Black patients is $\delta + \tau$, while the absolute effect for non-Black patients is δ.

Interpretation. The coefficient is not the effect of seeing a Black provider. It is the effect of being exposed to a local provider workforce with a higher Black-provider share, including direct and indirect channels.

3.3 Balance and selection tests

The paper estimates specifications that replace ΔY_{ij} with baseline patient characteristics or pre-move health measures. It also estimates a moving-incidence model:

$$Move_i = \alpha + \beta B_i + \delta P_j^B + \tau B_i \times P_j^B + \Omega P_j^X + \gamma B_i \times P_j^X + \pi M_i + \Psi B_i \times M_i + \varepsilon_{ij}. \quad (2)$$

Interpretation. These tests ask whether Black patients are differentially selected into moving away from or toward bases with different Black-provider shares. The results show little evidence of such selection.

3.4 Event-study specification

The event-study figures estimate period-specific versions of the core triple-difference coefficient:

$$\tau_k, \quad k \in \{\text{pre-move years, move year, post-move years}\}.$$

Interpretation. Pre-move coefficients test whether Black relative to non-Black patients were already trending differently before a move. Post-move coefficients show whether exposure accumulates with time.

3.5 Mechanism tests

The paper studies whether the results look like patient trust and communication rather than purely provider-driven treatment intensity.

- Medication adherence is decomposed into intensive and extensive margins.
- ER care is examined because emergency settings involve less patient discretion.
- Scheduled versus unscheduled caesareans are compared because planned deliveries require more patient-provider communication.
- Cancer screening is used as another preventive-care setting.

Interpretation. Strong effects in chronic and planned-care settings, but weak effects in ER care, point toward trust and communication channels.

3.6 Mortality decomposition

The paper decomposes the mortality effect using medical-literature estimates of the effect of medication fill days on mortality:

$$\begin{aligned}\text{Mortality effect from Table 2} &= \text{Medication fill-day effect from Table 2} && (3)\\ &\times \text{mortality effect of one fill-day increase} + \text{Residual.} && (4)\end{aligned}$$

Numerically:

$$-0.19 \approx 2.67 \times (-0.021) + \text{Residual.}$$

Interpretation. The paper concludes that about 31.1% of the mortality effect can be explained by improved medication adherence, while about 68.9% is residual.

4 Identification and empirical design

4.1 Why the MHS movers design is useful

- Military families often move across bases.
- Bases differ in provider racial composition.
- The model compares changes within movers, reducing bias from time-invariant patient differences.
- Sending and receiving base fixed effects absorb fixed base characteristics.
- Comparing Black and non-Black patients helps separate racial-disparity effects from general base-quality changes.

Interpretation. The identifying variation is not simply that some bases have more Black providers. It is that movers experience changes in Black-provider exposure, and those changes are compared across patient race.

4.2 Parallel-trends assumption

The key assumption is that, absent the provider-race change, Black-versus-non-Black outcome gaps would have evolved similarly for patients experiencing Black-provider-increasing and Black-provider-decreasing moves.

Interpretation. The paper supports this assumption with covariate balance, pre-move event studies, and tests showing no systematic relationship between moving and provider race.

4.3 Why base fixed effects matter

- Sending-base fixed effects control for the baseline environment from which patients depart.
- Receiving-base fixed effects control for destination-base characteristics.
- Including both means that identification comes from differential move-induced exposure by patient race, not from simple differences in base quality.

Interpretation. This helps address the concern that a base with more Black providers may also be different in other ways, such as patient demographics, provider age, or overall health-care quality.

4.4 Potential limitations

- Provider race is observed only for military-linked providers, not all purchased-care providers.
- The MHS population is not identical to the general U.S. population.
- Moves are plausibly exogenous in many respects but are not randomized.
- The paper estimates local effects for movers and chronic-disease patients.
- The mechanism is inferred through patterns across settings, not fully observed at the individual trust/communication level.

Interpretation. The paper is strongest as evidence on provider diversity in a large, insured, geographically mobile population. Generalizing to non-MHS settings requires care, although access effects outside the MHS could make diversity effects even larger.

5 Tables and figures

5.1 Table 1: Summary statistics

Read:

- The analytical sample contains 197,748 patient-by-disease-category observations.
- The average move-induced change in Black military provider share is close to zero, but the standard deviation is about 0.067.
- Roughly 24.5% of the analytical sample consists of Black patients.
- Average patient age is about 40, and 56.4% of patients are male.
- In the person-year chronic-disease sample, average medication fill days are about 76, and Black patients average about 72 fill days versus about 77 for non-Black patients.
- Systolic blood pressure and self-reported pain are higher among Black patients in the descriptive data.
- The average Black-provider share is 8.2% overall and is higher for Black patients' bases.

Interpretation. Table 1 provides two crucial pieces of context. First, the MHS chronic-disease sample is large enough to support a movers design. Second, there are meaningful baseline racial differences in preventive medication use and health indicators, making the question of reducing disparities substantive rather than purely symbolic.

Economic message. The descriptive statistics show both a treatment margin and a disparity margin: patients move across bases with different provider-race shares, and Black patients enter with lower preventive-care adherence and worse symptom/biomarker indicators.

What not to over-read. Table 1 is descriptive. The fact that Black patients are observed at bases with somewhat higher Black-provider shares does not identify the treatment effect; identification comes from move-induced changes and the triple-difference design.

TABLE 1
Summary statistics

	Full Sample		Black Patients		Non-Black Patients	
Variable	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
Panel A: Analytical sample. Unit of observation = person/disease category						
Move-induced change in Black military provider share	0.003	0.067	0.004	0.072	0.003	0.065
Incidence of mortality (at all over sample)	0.010	0.100	0.010	0.101	0.010	0.100
Patient age	39.98	11.15	39.88	10.49	40.06	11.35
Active-duty status	0.419	0.493	0.445	0.497	0.410	0.492
Male patient	0.564	0.498	0.560	0.496	0.567	0.496
Black patient	0.245	0.429	–	–	–	–
Incidence of junior enlisted pay grade	0.120	0.324	0.115	0.319	0.120	0.325
Incidence of senior enlisted pay grade	0.651	0.476	0.763	0.425	0.614	0.487
Incidence of junior officer pay grade	0.057	0.232	0.033	0.178	0.065	0.246
Incidence of senior officer pay grade	0.145	0.352	0.065	0.246	0.171	0.377
Panel B: Person-year chronic-disease sample (prior to creating move-induced changes)						
Annual medication fill days	76.024	134.228	72.428	129.284	76.984	135.502
Systolic level	130.699	11.763	132.354	11.900	130.089	11.652
Self-reported pain (1–10)	1.731	1.947	1.863	2.032	1.682	1.912
Black military provider share	0.082	0.066	0.095	0.070	0.078	0.065
Military provider age	41.567	3.044	41.907	2.983	41.465	3.054
Male military provider share	0.578	0.116	0.573	0.112	0.580	0.118
Share of black beneficiaries on base	0.193	0.075	0.222	0.083	0.184	0.070
Age of beneficiaries on base	29.926	3.543	30.019	3.614	29.899	3.522
Share of male beneficiaries on base	0.559	0.036	0.556	0.035	0.559	0.036

Notes: Data are from the Military Health System Data Repository 2003–2013. Descriptive statistics from Panel A are taken from a person-disease-category sample of chronic-disease patients (N = 197,748). Patient characteristics (*e.g.* race, age, sex, pay-grade, active-duty status) are of the patient's first year in the sample. Descriptive statistics from Panel B are taken from a person-year-disease-category sample of patients with four chronic diseases prior to collapsing down to the person-disease-category level. Both cases focus on patients that live within 10 miles of the base ($N = 1{,}886{,}984$). We assign each patient-year cell a base ($N = 116$) according to the MTF catchment/service area (designated by the MHS) associated with the closest MTF clinic to the patient's residence at that time. Distance to the base is signified by the distance between the zip-code centroid of the patient's residence and the zip-code centroid of the closest MTF clinic.

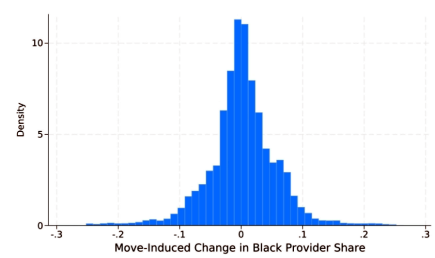

FIGURE 1
Distribution of move-induced changes in Black-provider shares
Notes: this figure presents a histogram of move-induced changes in prevailing shares of Black providers across a sample of chronic-disease patients who move over the sample period (N = 197748)

5.2 Figure 1: Distribution of move-induced changes in Black-provider shares

Read:

- The figure plots the distribution of changes in Black-provider share experienced by movers.
- There is support on both positive and negative changes, rather than all movers moving in one direction.

- The distribution has meaningful mass over large provider-share changes, including roughly between -20 and +20 percentage points.
- The identifying variation is therefore not a tiny mechanical change; many patients experience materially different provider racial environments after moving.

Interpretation. Figure 1 is a first-stage-type diagnostic. It shows the empirical variation behind the treatment variable. Without broad variation in move-induced provider diversity, the paper could not credibly estimate effects on disparities.

Economic message. The movers design is economically meaningful because moving can substantially change exposure to Black providers. The treatment is not just a statistical artifact.

What not to over-read. Figure 1 does not by itself show exogeneity. It only shows variation. Exogeneity is addressed later through balance, pre-trend, and robustness checks.

5.3 Table 2: Preventive care and mortality

Read:

- Panel A estimates effects on medication fill days.
- Without base fixed effects, the coefficient is 2.992 fill days; with sending and receiving base fixed effects, it is 2.676 fill days.
- Both estimates are statistically significant at the 1% level.
- Panel B estimates effects on mortality.
- The mortality coefficient is about -0.0018 to -0.0019, statistically significant at the 1% level.
- The paper interprets this as about a 0.2 percentage-point mortality reduction, roughly 18% relative to the mean mortality rate.

Interpretation. Table 2 is the central result. It says that Black patients gain relative to non-Black patients when they move to bases with a higher share of Black providers. The gains appear both in an intermediate input—preventive medication adherence—and in an ultimate outcome—mortality.

Economic message. Provider diversity appears to reduce racial health disparities through meaningful health-production channels. The mortality effect is especially important because it shows that the adherence gains are not merely administrative or billing artifacts.

What not to over-read. The estimate is a relative effect for Black versus non-Black patients. It should not be read as a universal mortality effect for all patients, nor as the isolated effect of one racially concordant visit.

5.4 Table 3: Biomarkers and patient-reported outcomes

Read:

- Panel A studies systolic blood pressure among diabetes and hypertension patients.
- The coefficient is about -0.326 without base fixed effects and -0.287 with base fixed effects.
- Panel B studies self-reported pain on a 0–10 scale.

TABLE 2

Relationship between increased racial diversity of providers and changes in racial health disparities, for preventive care and mortality

Panel A. Adherence analysis		
Dependent variable = change (post-move minus pre-move) in medication fill days.		
Black patient X 1-standard deviation change in base Black-provider share	2.992*** (1.007)	2.676*** (0.939)
N	130,952	130,952
Panel B. Mortality analysis		
Dependent variable = incidence of mortality over sample.		
Black patient X 1-standard deviation change in base Black-provider share	−0.0018*** (0.0007)	−0.0019*** (0.0007)
N	197,748	197,748
Pre- and post-move base fixed effects?	No	Yes

Notes: The unit of observation is a patient-by-disease-category cell. Standard errors are clustered at the sending-by-receiving base level. The sample includes adult MHS patients who consistently live within 10 miles of the base and who have moved once over the sample period. Specifications include fixed effects capturing the focal patient's incidence of each possible combination of the four disease categories, along with fixed effects capturing the disease category associated with the unit of observation. Coefficients of the Black-patient indicator and the move-induced change of the base Black-provider share are omitted for brevity (but see Supplementary Table A2 of the Online Appendix). Specifications include the additional controls included in Specification (1). Following the relevant guidelines, as discussed further in the Supplementary Online Appendix, the preventive-care analysis imposes further sample restrictions—*e.g.* only following medication adherence of hypertension patients who are not in control of their blood pressure. Nonetheless, as demonstrated in the Supplementary Online Appendix, the mortality results are virtually identical when imposing those medication-specific sample restrictions.
*Significant at the 10% level; ** Significant at the 5% level; ***Significant at the 1% level.

TABLE 3

Relationship between increased racial diversity of providers and changes in racial health disparities, for biomarkers and other outcomes

Panel A. Blood pressure analysis (diabetes and hypertension samples pooled)		
Dependent variable = change (post-move minus pre-move) in systolic level		
Black patient X 1-standard deviation change in base Black-provider share	−0.326** (0.164)	−0.287* (0.156)
N	43,456	43,456
Panel B. Self-reported pain (all samples pooled)		
Dependent variable = change (post-move minus pre-move) in self-reported pain (0–10)		
Black patient X 1-standard deviation change in base Black-provider share	−0.036 (0.024)	−0.055** (0.025)
N	87,075	87,075
Pre- and post-move base fixed effects?	No	Yes

Notes: These results are from specifications that follow those estimated in Table 2, except now focusing on the indicated dependent variables and on the indicated samples. These data are from the more limited set of vital statistics records in the MDR, which are for on-base records only and for 2009–2013 only.
*Significant at the 10% level; ** Significant at the 5% level; ***Significant at the 1% level.

- The pain coefficient becomes statistically significant with base fixed effects, at -0.055.
- These outcomes are measured in more limited vital-statistics data from on-base records during 2009–2013.

Interpretation. Table 3 is an outcome-validation table. It asks whether provider diversity affects clinical or symptom outcomes beyond claims-based medication fills. The negative blood-pressure and pain coefficients suggest that the care improvements have physiological and patient-experience consequences.

Economic message. The table strengthens the interpretation that provider diversity improves health management, not merely the volume of prescriptions.

What not to over-read. These are smaller and more limited samples than Table 2. The results are supportive, but the main evidence remains the larger medication and mortality analysis.

5.5 Table 4, Figure 2, and Figure 3: balance and pre-trends

Read:

- Table 4 replaces the main outcomes with baseline patient characteristics and pre-move health measures.
- Most coefficients are small and statistically insignificant.
- Predicted mortality based on observables is essentially unrelated to Black-provider-increasing moves, while the actual mortality effect in Table 2 is much larger.
- Figure 2 shows no systematic pre-move differential trends in patient characteristics for Black relative to non-Black patients.
- Figure 3 shows no pre-move increase in medication fill days before a Black-provider-increasing move.
- In Figure 3, medication fill days begin diverging after the move, consistent with exposure-driven effects.

Interpretation. These are the most important credibility diagnostics. If Black patients moving to more diverse-provider bases were already healthier, more adherent, or improving before the move, the main estimates would be less convincing. The balance and event-study evidence instead suggests that the relative improvement appears after exposure changes.

Economic message. The empirical design behaves like a plausible quasi-experiment: the treatment changes at the move, and the outcome response appears afterward rather than before.

What not to over-read. Balance tests cannot prove the absence of all unobserved confounding. They show that observable health, demographics, and pre-trends do not line up in a way that would naturally explain the main results.

TABLE 4

Covariate balance. Relationship between 1-standard-deviation move-induced increase in Black-provider share and differential in Black-versus-non-Black patient characteristics

Age (mean in pooled samples = 39.98)	−0.035 (0.169)	−0.126 (0.091)
Active-duty status (mean in pooled samples = 0.4192)	−0.003 (0.006)	0.002 (0.005)
Male (mean in pooled samples = 0.56)	−0.000 (0.004)	0.001 (0.004)
Annual RVU (mean in pooled samples = 51.38)	0.011 (0.505)	0.050 (0.476)
Charlson Index (mean in pooled samples = 0.47)	−0.006 (0.007)	−0.007 (0.007)
Inpatient days (mean in pooled samples = 0.78)	−0.033 (0.031)	−0.033 (0.031)
Incidence of junior enlisted pay grade (mean in pooled samples = 0.09)	0.005 (0.003)	0.004 (0.003)
Incidence of senior enlisted pay grade (mean in pooled samples = 0.68)	0.007 (0.007)	0.010* (0.005)
Incidence of junior officer pay grade (mean in pooled samples = 0.05)	−0.001 (0.002)	−0.001 (0.002)
Incidence of senior officer pay grade (mean in pooled samples = 0.15)	−0.010 (0.008)	−0.012 (0.005)
Predicted incidence of mortality (based on observable patient characteristics; mean in pooled samples = 0.010)	−0.00001 (0.00014)	−0.00007 (0.00011)
Pre-move medication fill days (mean in pooled samples = 54.3)	−0.240 (0.996)	−0.357 (0.909)
N	197,748	197,748
Pre- and post-move base fixed effects?	No	Yes

Notes: The specifications are identical to that estimate in Table 2 except that all baseline patient health characteristics are excluded. Instead, those characteristics are included as dependent variables in separate specifications estimated across each row. Covariate balance is conducted on the sample with the broadest eligibility criteria among those specifications estimated in Tables 2 and 3, though the results from this exercise generalize to focusing on the limited samples estimated in some specifications in Tables 2 and 3 based on the eligibility criteria for the relevant clinical context.
*Significant at the 10% level; ** Significant at the 5% level; ***Significant at the 1% level.

actual mortality, we estimate a τ coefficient of only −0.00007 using predicted mortality. With a 95% confidence interval for this latter estimate spanning from −0.00029 to 0.00015, our actual

Table 4: Covariate balance.

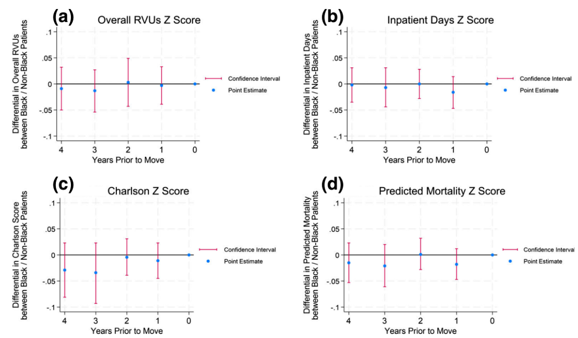

FIGURE 2

Relationship between differential in Z-scores of patient characteristics between Black and non-Black patients in years leading up to move associated with 1-standard deviation increase in Black provider share

Notes: each point estimate depicted is from a different regression, where the dependent variable represents the difference in levels for

Figure 2: Pre-move balance event studies.

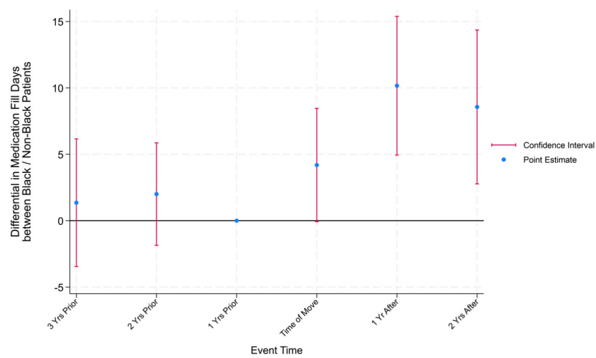

FIGURE 3

Event-study results. Relationship between Increased Racial Diversity of Providers and Changes in Racial Health Disparities in Prevention Adherence Measures in Years Leading up to and Following Patient Move

Notes: each point estimate depicted is from a different regression, where the dependent variable represents the difference in levels for the medication fill-day measure at the indicated event time relative to the reference period (the year prior to the move). The right-hand side of each regression follows that of specification (1) and the figure presents estimates of the τ interaction coefficients. Accordingly, this figure traces how medication fill days—for Black relative to non-Black patients—evolves in the time leading up to and following a move event, where a move is characterized by a 1-standard deviation increase in the prevailing Black-provider share. Specifications include sending and receiving base fixed effects. We explore a balanced sample and include only those patients we can observe over the entire event window and who consistently live within 10 miles of the base (N = 13,239)

Figure 3: Medication fill-day event study.

5.6 Table 5: Medication intensive and extensive margins

Read:

- Panel A repeats the main fill-day result from Table 2.
- Panel B restricts to patients with some medication fills in both pre- and post-move periods.
- The intensive-margin coefficient is about 6.3–6.9 fill days and is statistically significant.
- Panel C studies whether patients start having any fill days at all.
- Extensive-margin coefficients are small and statistically insignificant.

- Among patients with no fills before the move, the effect on starting any fills is also small.

Interpretation. Table 5 shows that the preventive-care effect is mostly about better adherence among patients already connected to medication treatment, not about newly initiating medication among those with no fills. This matters for mechanism: the provider-diversity effect seems to improve sustained chronic-disease management rather than simply causing first-time medication uptake.

Economic message. The main effect is an adherence-intensity effect. Provider diversity likely improves the ongoing relationship, trust, communication, or follow-through needed to maintain treatment.

What not to over-read. The absence of an extensive-margin effect does not mean provider diversity cannot improve initiation in other settings. It means that, in this chronic-disease sample and claims measurement framework, the estimated Table 2 result is driven primarily by intensive adherence.

TABLE 5

Medication intensive and extensive margin analysis

Panel A: Intensive and extensive margins collectively (primary results from Table 2)		
Dependent variable = change (post-move minus pre-move) in medication fill days		
Black patient X 1-standard deviation change in base Black-provider share	2.992*** (1.007)	2.676*** (0.939)
N	130,952	130,952
Panel B: Intensive margin		
Dependent variable = change (post-move minus pre-move) in medication fill days among those chronic-disease patients with some fills in both the pre- and post-move period (mean medication fills among this sample = 218.29)		
Black patient X 1-standard deviation change in base Black-provider share	6.936*** (2.599)	6.305*** (2.572)
N	58,186	58,186
Panel C: Extensive margin		
Dependent variable = change (post-move minus pre-move) in incidence of some medication fill days among all chronic-disease patients (mean incidence of some medication fill days = 0.29)		
Black patient X 1-standard deviation change in Base Black-provider share	0.006 (0.004)	0.005 (0.004)
N	130,952	130,952
Dependent variable = change (post-move minus pre-move) in incidence of some medication fill days among chronic-disease patients with no medication fills in the pre-move period (mean incidence of some medication fill days = 0.29)		
Black patient X 1-standard deviation change in base Black-provider share	0.003 (0.004)	0.003 (0.003)
N	92,064	92,064
Including fixed effects for pre- and post-move bases?	No	Yes

Notes: The specifications and samples estimated are identical to that estimated in Table 2 except that the samples and dependent variables are as indicated.
*Significant at the 10% level; ** Significant at the 5% level; ***Significant at the 1% level.

Table 5: Medication intensive and extensive margins.

5.7 Table 6 and Table 7: Mechanism evidence from ER care and childbirth

Read:

- Table 6 studies ER care, where patient discretion and long-run trust are less central.
- Provider diversity has no significant effect on average ER RVUs, diagnostic procedures, or mortality after ER visits.
- Table 7 studies scheduled versus unscheduled caesarean deliveries.
- Scheduled caesarean incidence rises by about 0.012 for Black relative to non-Black patients after Black-provider-increasing moves.
- Unscheduled caesarean incidence shows no meaningful effect.
- The planned versus unplanned contrast is important because scheduled caesareans involve more advance discussion and patient-provider decision-making.

Interpretation. These tables are mechanism tests. If the effect were mainly provider-side clinical intensity or general treatment bias, one might expect effects in emergency care or acute settings. The fact that effects appear in planned, relationship-dependent settings supports a trust and communication mechanism.

Economic message. Provider diversity matters most when patients have time to form expectations, communicate preferences, and comply with care plans. This is consistent with a demand-side or relational channel, not only a supply-side treatment-intensity channel.

What not to over-read. Table 6 does not prove that provider bias is irrelevant. It shows that the paper's main pattern is not broadly present in ER treatment intensity. Table 7 is suggestive because planned and unplanned caesareans differ in many ways, but the contrast is consistent with the paper's trust-based interpretation.

TABLE 6

Relationship between increased racial diversity of providers and changes in racial health disparities, for emergency room care

Panel A: Dependent variable = change (post-move minus pre-move) in average RVU total per ER admission (mean RVU per ER admission = 1.865)		
Black patient X 1-standard deviation change in base Black-provider share	−0.007 (0.009)	−0.010 (0.010)
N	470,677	129,929
Panel B: Dependent variable = change (post-move minus pre-move) in average number of diagnostic procedures per ER admission (mean number of diagnostic procedures per ER admission = 0.63)		
Black patient X 1-standard deviation change in base Black-provider share	−0.008 (0.010)	0.008 (0.016)
N	502,084	151,863
Panel C: Dependent variable = incidence of mortality in post-move period (mean mortality incidence over sample = 0.0096) (reported coefficients multiplied by 1,000)		
Black patient X 1-standard deviation change in base Black-provider share	0.366 (0.466)	0.155 (0.432)
N	502,269	151,916
Sample restricted to diagnosis codes with weekend shares ≥ 0.24	No	Yes

Notes: This table presents results from a sample of patients that move across bases once during the sample period and that visit the emergency room in each of the pre- and post-move periods. The specifications estimated are otherwise structured similarly to those estimated in Table 2. The move-induced change in the Black-provider share (based on providers who see patients in the emergency room) is normalized such that coefficients capture the effect of a 1-standard deviation change in the Black-provider share (1-standard deviation represents 0.11).
*Significant at the 10% level; ** Significant at the 5% level; ***Significant at the 1% level.

ER visits with diagnosis codes with large weekend shares, suggesting non-discretionary visits), and the results hold. The lack of provider diversity effects on the delivery of ER care, when they are so prevalent for chronic care, further bolsters the suggestion that patient trust is a meaningful

Table 6: Emergency-room care.

TABLE 7

Relationship between increased racial diversity of providers and changes in racial health disparities for caesarean section delivery incidence, scheduled, and unscheduled

Panel A: Dependent variable = change (post-move minus pre-move) in incidence of scheduled caesarean delivery (mean incidence of scheduled caesarean delivery = 0.190)		
Black patient X 1-standard deviation change in base Black-provider share	0.012* (0.007)	0.012* (0.007)
N	37,304	37,304
Panel B: Dependent variable = change (post-move minus pre-move) in incidence of unscheduled caesarean delivery (mean incidence of unscheduled caesarean delivery = 0.061)		
Black patient X 1-standard deviation change in base Black-provider share	−0.004 (0.006)	−0.003 (0.006)
N	37,304	37,304
Pre- and post-move base fixed effects?	No	Yes

Notes: This sample includes mothers who have delivered more than once and moved across bases between the moves, where both the pre- and post-move base has a base hospital. The specification is otherwise structured similarly to that estimated in Table 2, except that the specifications also include the following unavoidable delivery risk factors (following Frakes *et al*., 2023a): breech presentation, multiple deliveries, cord prolapse, placenta previa and placenta abruption. The move-induced change in the Black-provider share (based on providers who deliver children on the base) is normalized such that coefficients capture the effect of a 1-standard deviation change in the Black-provider share (1-standard deviation represents 0.10).
*Significant at the 10% level; ** Significant at the 5% level; ***Significant at the 1% level.

Table 7: Caesarean delivery incidence.

5.8 Table 8 and mortality decomposition: cancer screening and preventive-care mechanism

Read:

- Table 8 studies cancer-screening guideline compliance, pooling cervical and breast cancer screening samples.
- A one-standard-deviation increase in the Black-provider share increases screening compliance for Black relative to non-Black patients by 0.006–0.009.
- The effect is statistically significant and represents roughly 0.8–1.3% relative to the mean screening rate.
- The mortality decomposition uses Table 2 medication fill-day effects and medical-literature estimates linking preventive medication adherence to mortality.
- About 31.1% of the mortality effect is attributed to improved medication fill days.
- The remaining 68.9% is residual and could reflect other preventive behaviors, lifestyle, monitoring, trust, diet, exercise, or unmeasured care quality.

Interpretation. Table 8 extends the main logic beyond chronic medication adherence into another preventive-care domain. The decomposition then links adherence to mortality, showing that medication use plausibly explains a meaningful but incomplete share of the survival effects.

Economic message. Provider diversity appears to operate through a broad preventive-care channel. Medication adherence is important, but the mortality result is probably not explained by a single observed behavior.

What not to over-read. The mortality decomposition is not a causal mediation design. It combines the paper's estimates with external medical-literature effects, so the 31.1% figure should be read as an approximate accounting exercise rather than a definitive mediation estimate.

TABLE 8

Relationship between increased racial diversity of providers and changes in racial health disparities in cancer screening rates (pooling over cervical cancer and breast cancer samples)

Black patient X 1-standard deviation change in base Black-provider share	0.009** (0.004)	0.006* (0.003)
N	143,701	143,701
Including fixed effects for pre- and post-move bases?	No	Yes
Mean cancer-screening rate	0.69	

Notes: The results presented are from a sample of women between the ages of 20 and 65 who move across bases over the sample (cervical cancer sample) pooled with a sample of women between the ages of 50 and 65 who likewise move across bases over the sample (breast cancer sample). The results presented are the estimated τ coefficients from specification (1) on this pooled sample. The regression includes a dummy indicating the relevant sub-sample. Standard errors are clustered at the sending-by-receiving base level. The move-induced change in the Black-provider share is normalized such that coefficients capture the effect of a 1-standard deviation change in the Black-provider share (1-standard deviation represents 0.12).

*Significant at the 10% level; **Significant at the 5% level; ***Significant at the 1% level.

Table 8: Provider diversity and cancer-screening disparities.

6 Short summary

- The paper estimates whether increasing the Black-provider share reduces racial health disparities.
- The setting is the Military Health System, which has claims data, mortality data, provider demographics, and large geographic scope.
- Provider race is identified through providers' beneficiary records in the MHS.
- The empirical design uses patient moves across military bases and variation in base-level provider racial composition.
- The key coefficient is a triple-difference interaction: Black patient $\times$ move-induced change in Black-provider share.
- A one-standard-deviation increase in Black-provider share raises medication fill days for Black relative to non-Black patients by roughly 2.7–3.0 days.
- The same provider-diversity increase reduces mortality by about 0.18–0.19 percentage points.
- Biomarker and pain results support the interpretation that health improves, not just prescription volume.
- Covariate balance and pre-trend tests show little evidence that Black-provider-increasing moves select healthier or improving Black patients.
- The medication result is mostly intensive margin: better adherence among patients already taking medication.
- ER-care results show no strong effects, suggesting the main channel is not general provider-side treatment intensity.
- Scheduled caesarean and cancer-screening results support the importance of patient trust, communication, and preventive decision-making.
- The authors argue that spillovers are important and that studying only direct patient-provider concordance understates the value of workforce diversity.
- Improved medication adherence explains roughly 31% of the mortality effect.
- The broader conclusion is that provider diversity can be an input into reducing racial health disparities.

7 Conclusion

7.1 Core conclusion

Frakes and Gruber provide evidence that increasing the racial diversity of the provider workforce can reduce racial health disparities. The result is not limited to a narrow physician-patient match. Instead, the paper estimates the effect of a more racially diverse provider environment and finds improvements in medication adherence, biomarkers, pain, cancer screening, and mortality for Black relative to non-Black patients.

7.2 Economic intuition

The economic intuition is that medical care is not only a technical input delivered by providers. It is also a relationship-intensive service. In chronic disease, patients must trust the medical system, understand treatment instructions, adhere to medications, return for monitoring, and update lifestyle choices. A more diverse provider workforce can improve this process directly through concordant interactions and indirectly through spillovers to non-Black providers and patient trust in the local medical system.

7.3 Why the setting matters

The MHS setting is especially useful because it provides a large insured population, repeated claims, mortality outcomes, provider race, and cross-base moves. In many non-MHS settings, lack of insurance or lack of access would complicate interpretation. Here, the paper can focus more directly on the role of provider diversity conditional on broad access to care.

7.4 Contribution relative to concordance studies

Many prior studies estimate the effect of racial concordance in a single encounter. This paper estimates a more policy-relevant parameter: what happens when the local provider workforce becomes more racially diverse. The distinction matters because workforce diversity can change patient sorting, provider learning, peer effects, institutional trust, and spillovers. The estimated effect can therefore be larger and broader than the sum of individual concordant visits.

7.5 Interpretation of the mechanism

The evidence points most strongly toward trust and communication. Effects are concentrated in chronic care, medication adherence, screening, and scheduled caesarean decisions—settings where patients make decisions over time and where communication matters. Effects are weak in ER care, where immediate provider decisions dominate and patient discretion is more limited.

7.6 Policy implications

The results suggest that policies affecting provider diversity—medical school admissions, residency pipelines, military and public-sector hiring, provider placement, and retention of minority providers—may have health-disparity consequences. The findings also suggest that provider diversity can matter even in systems with universal coverage, meaning that insurance access alone may not be sufficient to eliminate racial health disparities.

7.7 Limits and caution

The paper does not imply that provider diversity is the only or even the dominant solution to racial health disparities. Nor does it prove that every setting will produce the same magnitude of effect. The estimates are based on movers within the MHS and on chronic-disease patients. However, the strength of the mortality, adherence, and diagnostic evidence makes the paper a compelling demonstration that workforce composition can affect health production.

7.8 Final takeaway

The main takeaway is that the racial composition of providers is not merely a distributional or representational issue. It can affect measurable health outcomes. In a setting where patients are insured and care is available, increasing provider diversity still reduces racial disparities, implying that trust, communication, and institutional representation are central components of effective health care.